

class09

Importing candy data

```
candy_file <- "candy-data.csv"
# Importing candy data
candy = read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

What is in the dataset?

Q1: How many different candy types are in this dataset?

85 different candy types.

```
# Number of candy types  
nrow(candy)
```

```
[1] 85
```

Q2: How many fruity candy types are in the dataset?

38 fruity candy types.

```
# Number of fruity candy types  
sum(candy$fruity)
```

```
[1] 38
```

What is your favorite candy?

Q3: What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

My favorite candy in the dataset is Skittles original. Its winpercent value is 63.08514

```
# winpercent value of "Skittles original"  
candy["Skittles original", ]$winpercent
```

```
[1] 63.08514
```

Q4. What is the winpercent value for “Kit Kat”?

The winpercent value for “Kit Kat” is 76.7686.

```
# winpercent value for "Kit Kat"  
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

The winpercent value for “Tootsie Roll Snack Bars” is 49.6535.

```
# winpercent value for "Tootsie Roll Snack Bars"
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

The `skimr::skim()` function

```
# using skim() for the candy dataset
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

Yes, the hist variable.

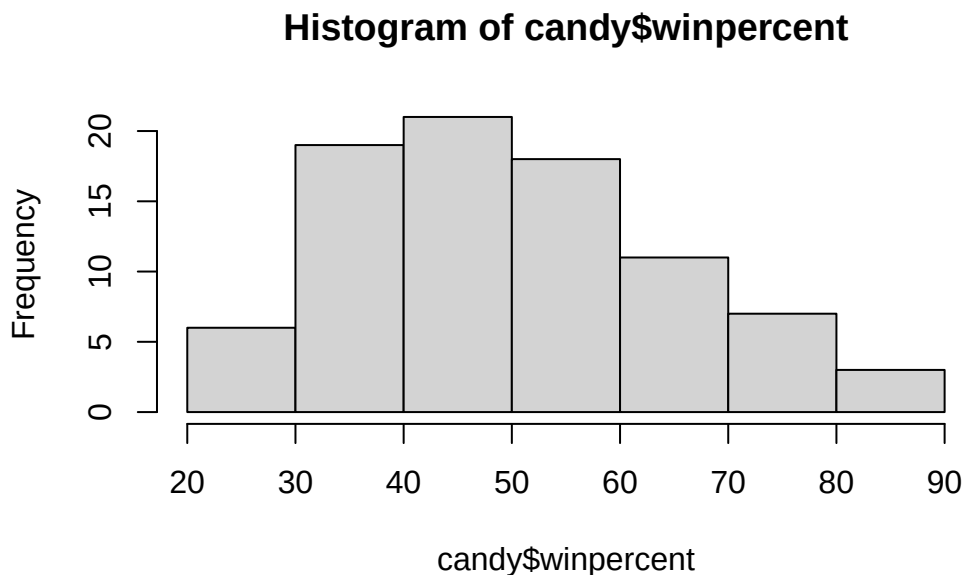
Q7. What do you think a zero and one represent for the candy\$chocolate column?

A one represents that all of the rows for that column have data, while a zero represents the opposite.

Exploratory analysis

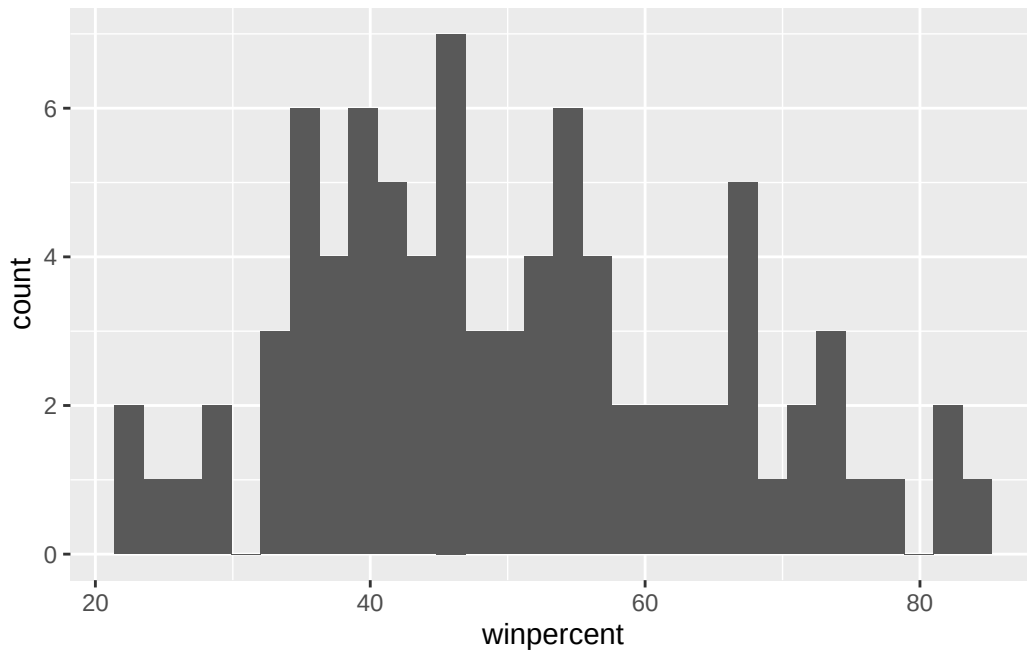
Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
# Base R histogram
hist(candy$winpercent)
```



```
# ggplot histogram
library(ggplot2)
ggplot(candy, aes(winpercent)) + geom_histogram()
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



Q9. Is the distribution of winpercent values symmetrical?

No the distribution of winpercent values is asymmetrical and is positively skewed.

Q10. Is the center of the distribution above or below 50%?

The center of the distribution is below 50%.

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

On average, chocolate candy is higher ranked than lower candy.

```
# average winpercent for chocolate candy
chocolate <- candy$winpercent[as.logical(candy$chocolate)]
mean(chocolate)
```

```
[1] 60.92153
```

```
# average winpercent for fruity candy
fruity <- candy$winpercent[as.logical(candy$fruity)]
mean(fruity)
```

```
[1] 44.11974
```

Q12. Is this difference statistically significant?

Yes, the difference is statistically significant, as seen by the t value being outside of the 95 percent confidence interval.

```
# Testing statistical significance of chocolate vs. fruity candy rankings
t.test(chocolate, fruity)
```

Welch Two Sample t-test

```
data: chocolate and fruity
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Overall candy rankings

Q13. What are the five least liked candy types in this set?

The five least liked candy types are “Nik L Nip”, “Boston Baked Beans”, “Chiclets”, “Super Bubble”, and “Jawbusters”

```
# Five least liked candy types
head(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat
Nik L Nip	0	1	0	0	0
Boston Baked Beans	0	0	0	1	0
Chiclets	0	1	0	0	0

Super Bubble	0	1	0	0	0	
Jawbusters	0	1	0	0	0	
	crisped	ricewafer	hard bar	pluribus	sugarpercent	pricepercent
Nik L Nip	0	0	0	1	0.197	0.976
Boston Baked Beans	0	0	0	1	0.313	0.511
Chiclets	0	0	0	1	0.046	0.325
Super Bubble	0	0	0	0	0.162	0.116
Jawbusters	0	1	0	1	0.093	0.511
	winpercent					
Nik L Nip	22.44534					
Boston Baked Beans	23.41782					
Chiclets	24.52499					
Super Bubble	27.30386					
Jawbusters	28.12744					

Q14. What are the top 5 all time favorite candy types out of this set?

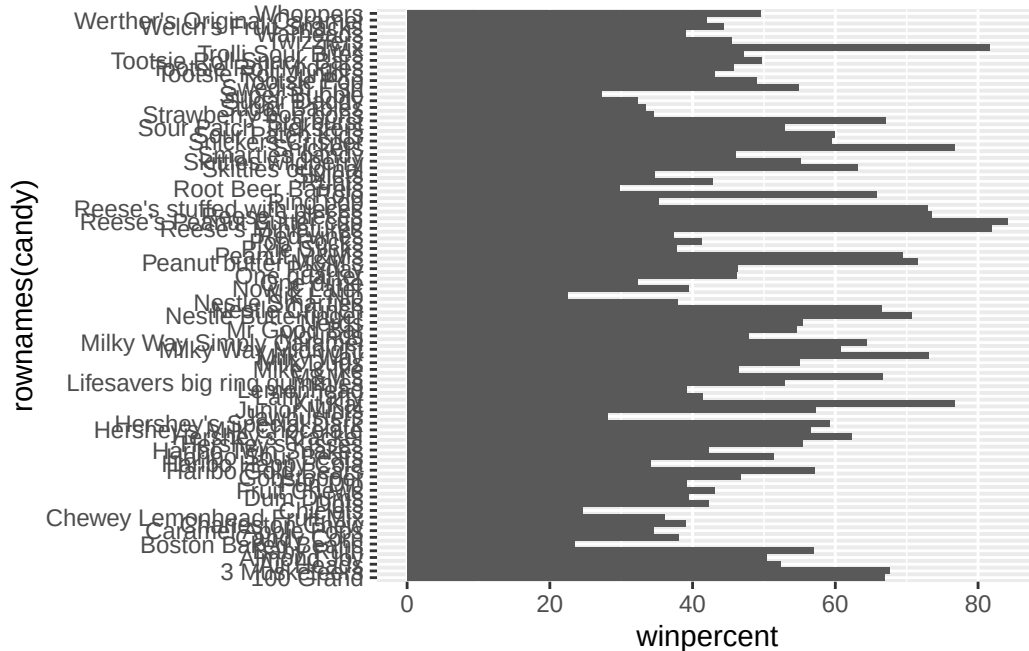
The top 5 all time favorite candy types are “Snickers”, “Kit Kat”, “Twix”, “Reese’s Miniatures”, and “Reese’s Peanut Butter cup”

```
# Five favorite candy types
tail(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Snickers	1	0	1		1	1
Kit Kat	1	0	0		0	0
Twix	1	0	1		0	0
Reese's Miniatures	1	0	0		1	0
Reese's Peanut Butter cup	1	0	0		1	0
	crisped	ricewafer	hard bar	pluribus	sugarpercent	
Snickers		0	0	1	0	0.546
Kit Kat		1	0	1	0	0.313
Twix		1	0	1	0	0.546
Reese's Miniatures		0	0	0	0	0.034
Reese's Peanut Butter cup		0	0	0	0	0.720
	pricepercent	winpercent				
Snickers	0.651	76.67378				
Kit Kat	0.511	76.76860				
Twix	0.906	81.64291				
Reese's Miniatures	0.279	81.86626				
Reese's Peanut Butter cup	0.651	84.18029				

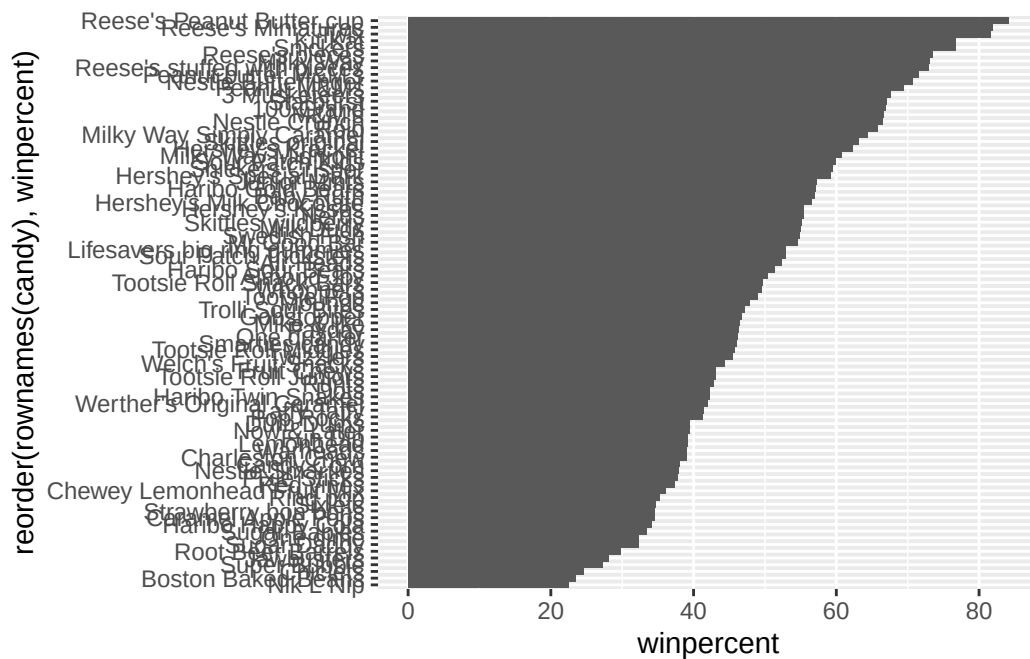
Q15. Make a first barplot of candy ranking based on winpercent values.

```
# Creating an initial winpercent barplot
ggplot(candy) + aes(winpercent, rownames(candy)) + geom_col()
```



Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

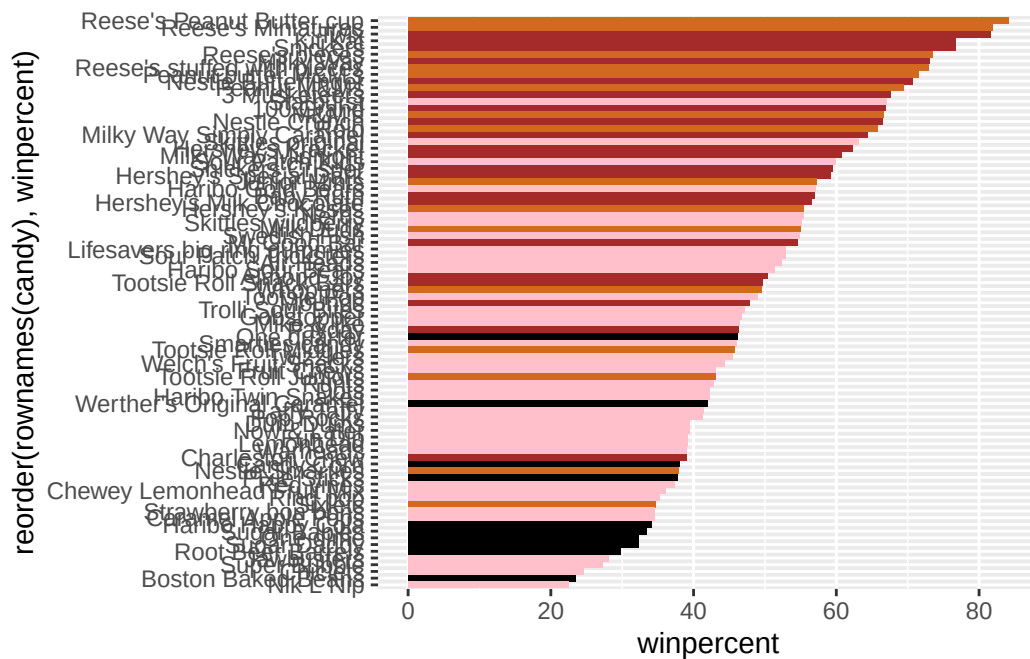
```
ggplot(candy) + aes(winpercent,
# Reordering the bars to sort by winpercent
reorder(rownames(candy), winpercent)) + geom_col()
```

Time to add some useful color

```
# Setting up a color vector
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"

# Plotting with these color vectors
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)
```



Q17. What is the worst ranked chocolate candy?

The worst ranked chocolate candy is Sixlets.

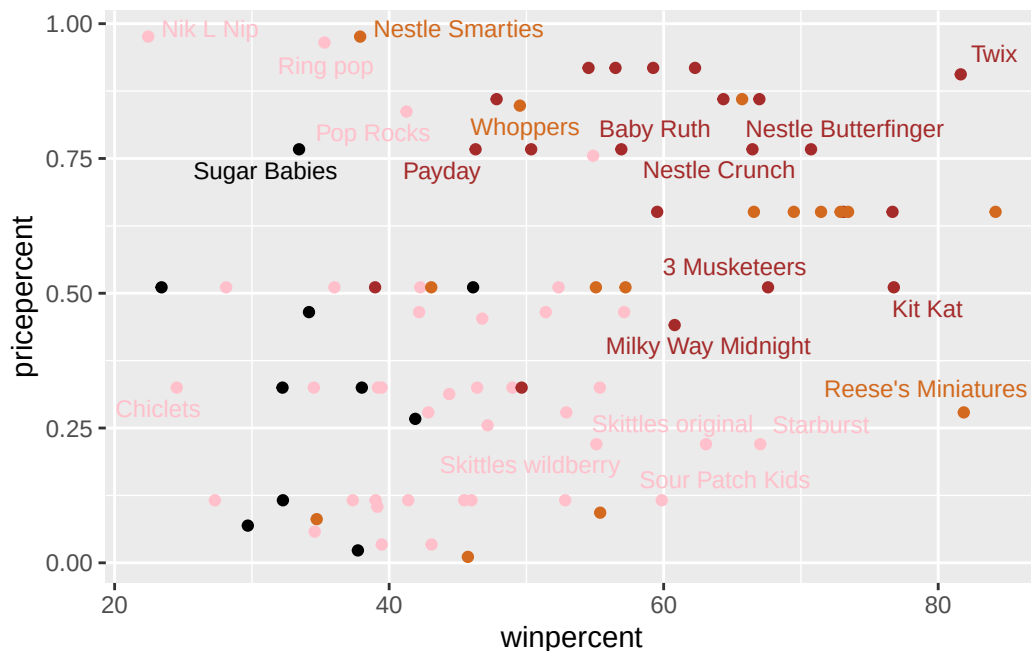
Q18. What is the best ranked fruity candy?

The best ranked fruity candy is Starburst.

Taking a look at pricepercent

```
library(ggrepel)

# How about a plot of win vs price
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=3.3, max.overlaps = 5)
```



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures are the highest ranked for the least money.

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

The top 5 most expensive candy types are Nik L Nip, Nestle Smarties, Ring pop, Hershey's Krackel, and Hershey's Milk Chocolate. Of these, the least popular is Nik L Nip.

```
# Ordering by price
ord <- order(candy$pricepercent, decreasing = TRUE)
head(candy[ord,c(11,12)], n=5 )
```

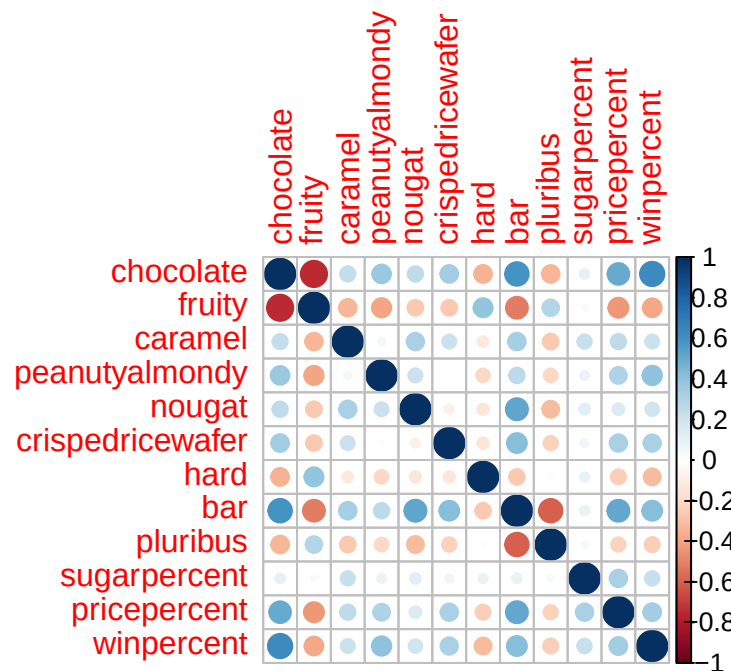
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
# Plotting a correlation matrix  
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Chocolate and fruity are anti-correlated.

Q23. Similarly, what two variables are most positively correlated?

Variables that align with themselves are the most positively correlated (i.e. chocolate and chocolate).

Principal Component Analysis

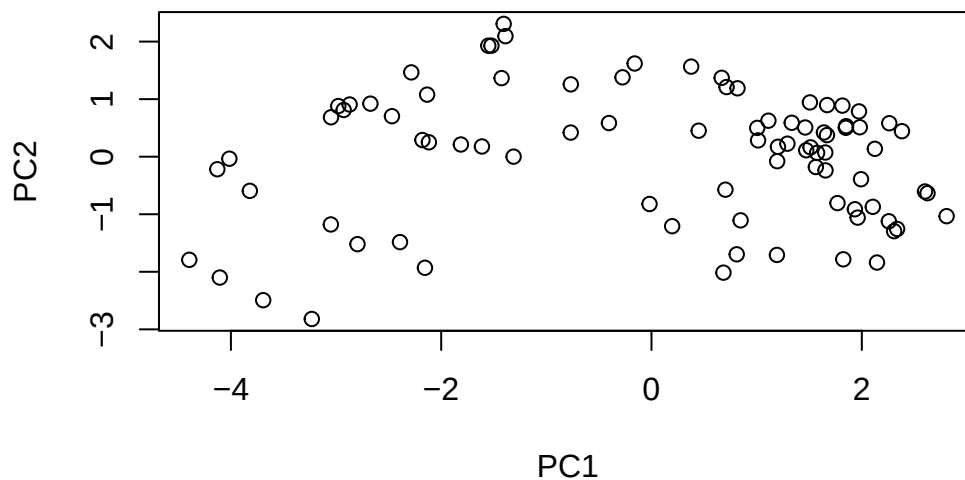
```
# PCA for the candy dataset
pca <- prcomp(candy, scale=TRUE)
summary(pca)
```

Importance of components:

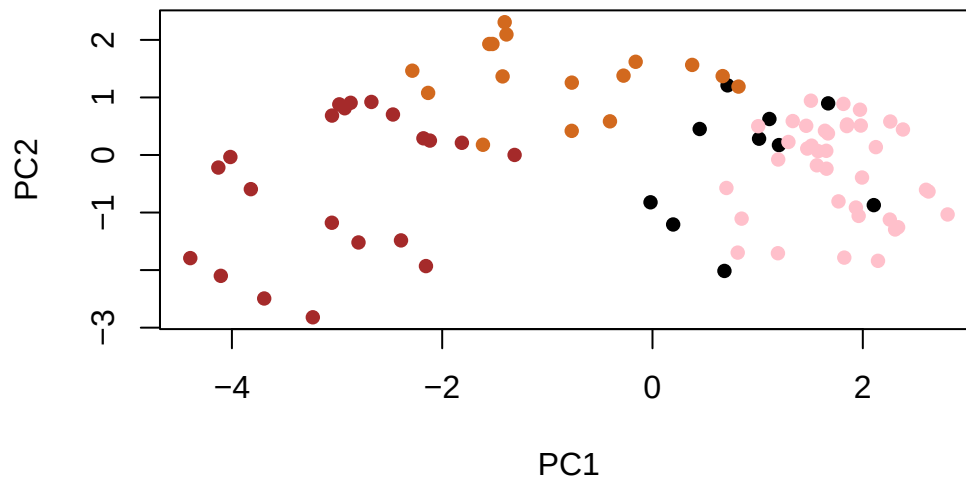
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
# Plotting PC1 vs PC2
plot(pca$x[,1:2])
```



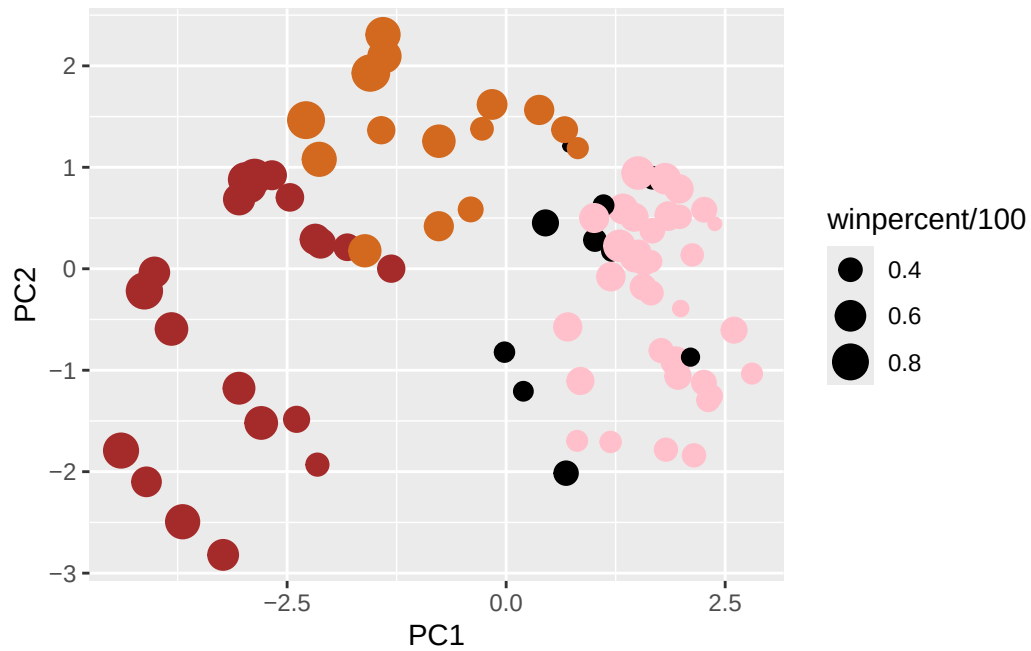
```
plot(pca$x[,1:2],
     # adding color and changing plotting character
     col=my_cols, pch=16)
```



```
# Make a new data-frame with our PCA results and candy data
my_data <- cbind(candy, pca$x[,1:3])
```

```
# Plotting in ggplot
p <- ggplot(my_data) +
  aes(x=PC1, y=PC2,
      size=winpercent/100,
      text=rownames(my_data),
      label=rownames(my_data)) +
  geom_point(col=my_cols)
```

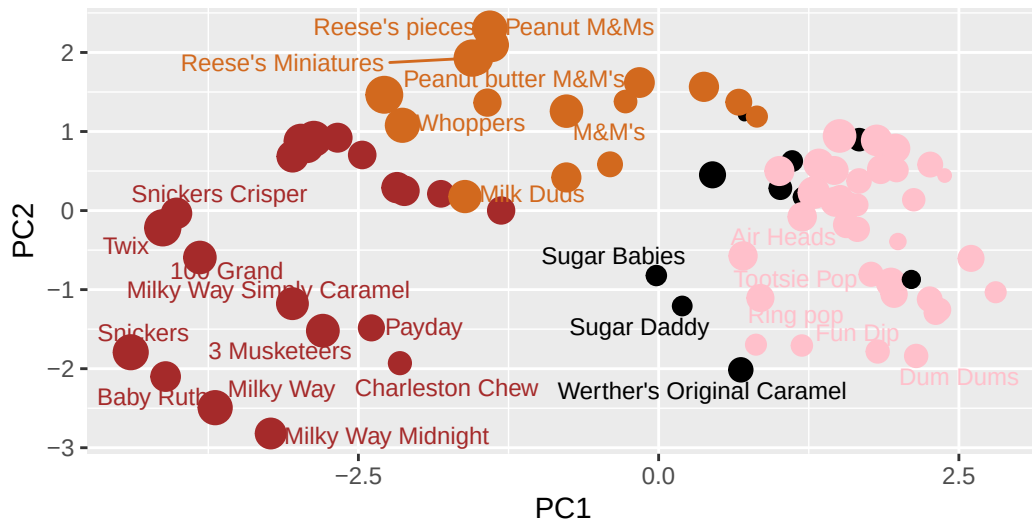
p



```
# Labeling the plot
p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
        caption="Data from 538")
```

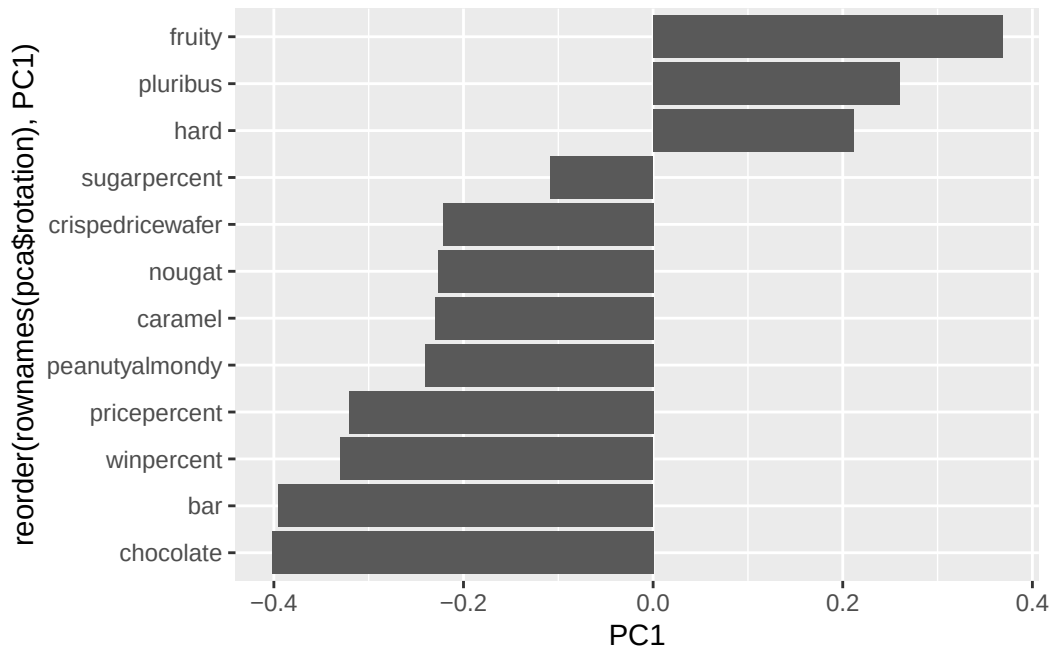
Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
# Bar plot of loadings
ggplot(pca$rotation) +
  aes(PC1, reorder(rownames(pca$rotation), PC1)) +
  geom_col()
```

Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

The “fruity”, “pluribus”, and “hard” variables are picked up strongly by PC1 in the positive direction, which makes sense because the “fruity” variable is on the positive side of the PC1 vs PC2 point plots, showing that it has the largest effect on PC1 in the positive direction.

Summary

Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a “winning” candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Chocolate, bar, peanuty/almondy, and candies with a lower price point tend to make a winning candy. The correlation matrix in relation to the winpercent variable complements PCA in terms of how important chocolate is to making a winning candy.